

# Program Design Document

## Program Name

GPACalc.cpp

## Purpose

Prompts the user to enter credits and grade and then calculates grades

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## Inputs

List all inputs the program will receive.

- Grade
  - Credit
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## Outputs

List what the program will display or produce.

Example:

- GPA
  - Class taken/Counter
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## Variables

| Variable    | Type   | Purpose                                            |
|-------------|--------|----------------------------------------------------|
| creditGrade | string | Stores the user's combined input (credits + grade) |
| counter     | int    | Counts the number of valid classes entered         |
| score       | double | Stores the GPA value of the current grade          |

|                           |                     |                                       |
|---------------------------|---------------------|---------------------------------------|
| <code>finalCredits</code> | <code>int</code>    | Stores the total number of credits    |
| <code>finalPoints</code>  | <code>double</code> | Stores total grade points accumulated |
| <code>grade</code>        | <code>string</code> | Stores the extracted letter grade     |

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## Key Design Choices

Explain any important decisions you made when designing the program.

- I used a while loop to continuously accept user input until the user enters "quit", allowing flexibility in the number of classes entered.
  - I used if-else statements to map letter grades to GPA values because it is straightforward and easy to read for multiple conditions.
  - I used a string input format instead of separate inputs to simplify user interaction.
  - I converted grades to uppercase to make the program case insensitive.
  - I validated credits to ensure they are between 1 and 9, preventing invalid calculations.
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## Program Steps (Algorithm)

1. Prompt the user to enter class credits and grade.
  2. Read user input.
  3. Check if the user entered "quit" → if yes, exit loop.
  4. Extract credits and grade from input.
  5. Validate credits (must be between 1 and 9).
  6. Convert grade to uppercase.
  7. Determine GPA value based on grade.
  8. If invalid grade, display error and continue.
  9. Multiply credits by grade points and add to total.
  10. Update total credits and class counter.
  11. Repeat until user exits.
  12. Calculate final GPA.
  13. Display total classes and GPA (rounded).
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## Functions

N/A

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## Sample Input/Output

Input:

3A

4B+

2C

quit

Output:

Classes taken: 3, Semester GPA: 3.26

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## Testing

### Test Case 1

Input: 3A, 4B+, 2C, quit

Expected Result: Correct GPA calculation

### Test Case 2

Input: quit

Expected Result: "No classes entered."

### Test Case 3

Input: 0A

Expected Result: "Invalid credits"

### Test Case 4

Input: 3Z

Expected Result: "Invalid grade"

### Test Case 5

Input: 3a, 4b+

Expected Result: Correct GPA (case-insensitive handling)

# AI Usage Reflection

1. Did you use any AI tools for this assignment? If so, which ones?

Yes, ChatGPT

2. How did you use it? (e.g., "I asked Claude to explain what a segmentation fault is after getting one I couldn't understand")

I used it for parsing and the rounding to the tenth

3. How did it help, or how did it fall short?

It definitely helped

4. What did you learn from the interaction that you applied yourself?

On how it can separate string into 2 variables